

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME: **ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01**
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)

Assessment Tool Weightage: 30%

Total Marks: 100

Time: 90 Mins

No. of Questions: 1

Annexure A Research Review & Analysis

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	

Total	100%	
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Signature of the Faculty

Total Marks Awarded

S.No.	Research Review & Analysis Question	Primary Topic
1	Decision Tree Learning: Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.	Decision Trees

Research Review & Analysis

Comparative Study of Decision Tree Learning: ID3, C4.5 and CART

Abstract

Decision trees are supervised learning methods used mainly for classification and prediction. This review compares ID3, C4.5, and CART based on attribute selection, pruning, interpretability, computational complexity, and performance. ID3 uses Information Gain, C4.5 uses Gain Ratio, and CART uses Gini Index. The review identifies limitations of traditional decision trees and proposes an adaptive approach to improve accuracy and interpretability. The structure follows the assessment requirements in the provided document.

1. Introduction

Decision trees are widely used supervised learning techniques. They represent decisions using a tree structure consisting of root nodes, internal nodes, branches, and leaf nodes.

They are popular because their decision rules are easy to understand. However, excessive tree growth can cause overfitting. ID3, C4.5, and CART are important decision-tree algorithms developed to address classification and prediction problems.

2. Research Problem and Objectives

Research Problem

ID3, C4.5, and CART use different attribute-selection and pruning methods. These differences affect their accuracy, complexity, interpretability, and generalization.

Objectives

1. Compare ID3, C4.5, and CART.

2. Analyze their attribute-selection methods.
3. Compare their pruning strategies.
4. Evaluate complexity and interpretability.
5. Identify research gaps.
6. Propose an improvement.

3. Literature Review

ID3

ID3 uses Entropy and Information Gain to select the best attribute. It is simple and mainly suited to categorical data.

Limitation: Information Gain can favor attributes having many distinct values.

C4.5

C4.5 is an improvement of ID3. It uses Gain Ratio, supports continuous attributes and missing values, and includes pruning.

Advantage: Better flexibility and overfitting control than basic ID3.

CART

CART stands for Classification and Regression Trees. It uses Gini Index for classification and creates binary trees. It also supports regression and cost-complexity pruning.

4. Comparative Analysis

Parameter	ID3	C4.5	CART
Attribute selection	Information Gain	Gain Ratio	Gini Index
Tree structure	Multiway	Multiway	Binary
Continuous data	Limited	Supported	Supported
Missing values	Limited	Supported	Supported
Pruning	Limited	Yes	Cost-complexity
Classification	Yes	Yes	Yes
Regression	No	No	Yes
Interpretability	High	High	High
Complexity	Low	Moderate	Moderate

Overall: ID3 is simplest, C4.5 is more flexible, and CART is more general-purpose.

5. Critical Analysis

ID3

Strength: Simple and easy to understand.

Weakness: Can overfit and has bias toward multi-valued attributes.

C4.5

Strength: Handles continuous/missing data and provides pruning.

Weakness: More computationally complex than ID3.

CART

Strength: Supports classification and regression with effective pruning.

Weakness: Binary splitting may produce deeper trees.

The main challenge for all three methods is balancing accuracy with tree complexity.

6. Research Gap

The major research gaps are:

1. **Greedy splitting:** The best local split may not produce the best overall tree.
2. **Accuracy vs. interpretability:** More complex trees can be harder to understand.
3. **Tree instability:** Small data changes may produce different tree structures.
4. **Fixed criteria:** A single splitting criterion may not be optimal for every dataset.

7. Proposed Research Direction

Adaptive Decision Tree

A possible improvement is an **Adaptive Decision Tree** that:

- Evaluates multiple splitting criteria.
- Selects the most suitable criterion for the dataset.
- Considers multiple candidate splits.
- Uses adaptive pruning.
- Controls tree depth and complexity.

Proposed flow:

Dataset → Preprocessing → Adaptive Split Selection → Tree Construction → Pruning → Evaluation

8. Proposed Research Methodology / Analysis Framework

1. Select datasets such as **Iris, Wine, and Breast Cancer**.
2. Preprocess the datasets.
3. Train ID3, C4.5, and CART.
4. Train the proposed adaptive model.
5. Compare the models using:
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - Training time
 - Tree depth
 - Number of nodes
6. Apply cross-validation for reliable evaluation.

9. Experimental / Evidence-Based Validation

The same datasets and preprocessing methods should be used for all algorithms.

The experiment should:

- Train each model.
- Test on unseen data.
- Calculate performance metrics.
- Measure tree complexity.
- Compare training time.
- Evaluate generalization using cross-validation.

This provides objective evidence for determining whether the proposed method improves existing approaches.

10. Result Analysis and Interpretation

The results should compare **accuracy and complexity together**.

Model	Accuracy	Tree Complexity	Training Cost
ID3	To be measured	To be measured	Low
C4.5	To be measured	To be measured	Moderate

Model	Accuracy	Tree Complexity	Training Cost
CART	To be measured	To be measured	Moderate
Proposed Model	To be measured	To be measured	To be measured

Note: Numerical results must be obtained from the actual experiment and should not be fabricated.

11. Discussion

ID3 provides a simple foundation for decision-tree learning. C4.5 improves ID3 through Gain Ratio, better data handling, and pruning. CART provides a flexible framework for both classification and regression.

The comparison shows that there is **no universally best algorithm**. The appropriate method depends on the dataset, required accuracy, computational resources, and interpretability requirements.

12. Conclusion

ID3, C4.5, and CART are important decision-tree learning algorithms with different strengths. **ID3 is simple, C4.5 is flexible, and CART is suitable for both classification and regression.**

The major research gap is achieving a balance between **accuracy, interpretability, stability, and computational efficiency**.

An adaptive decision-tree approach using improved split selection and pruning is a promising direction for future research.

13. References

1. Mienye, I. D., & Jere, N. (2024). **A Survey of Decision Trees: Concepts, Algorithms, and Applications.** *IEEE Access*, 12, 86716–86727.
2. Abedinia, A., & Seydi, V. (2024). **Building semi-supervised decision trees with semi-CART algorithm.** *International Journal of Machine Learning and Cybernetics*, 15, 4493–4510.
3. Chaouki, A., Read, J., & Bifet, A. (2024). **Online Learning of Decision Trees with Thompson Sampling.** *Proceedings of AISTATS 2024*, 2944–2952.
4. Bressan, M., & Sozio, M. (2024). **Fully-Dynamic Approximate Decision Trees With Worst-Case Update Time Guarantees.** *Proceedings of ICML 2024*, 4517–4541.

Structure of the Report:

S.No.	Paper Section	Expected Content
1	Title	Specific and concise title related to the selected research question/topic.
2	Abstract	Brief summary of the research problem, review methodology, major findings, research gap, and conclusion.
3	Keywords	4–6 important keywords related to the selected topic.
4	1. Introduction	Introduce the AI topic, background, importance, motivation, and scope of the review.
5	2. Research Problem and Objectives	Clearly state the research problem and specific objectives of the review.
6	3. Literature Review	Review relevant research papers and summarize major approaches, methodologies, findings, and contributions.
7	4. Comparative Analysis	Compare the reviewed approaches based on suitable parameters such as accuracy, complexity, interpretability, computational cost, scalability, and generalization.
8	5. Critical Analysis	Discuss strengths, weaknesses, limitations, assumptions, and practical challenges identified across the reviewed studies.
9	6. Research Gap	Identify unresolved problems, limitations in existing research, and areas requiring further investigation.
10	7. Proposed Research Direction	Suggest possible solutions, improvements, or future research directions based on the identified gap.
11	8. Discussion	Discuss the overall insights obtained from the review and their implications for AI research and applications.
12	9. Conclusion	Summarize the major findings, key insights, research gap, and future direction.
13	References	List all research papers, books, datasets, websites, and other sources using a consistent citation style.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.
4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.

5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.
7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.

10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.
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